$\boxed{\text{properties of the integral of a bounded fun}}$

0.1.1 f bounded on $[a,b]$

$\Leftrightarrow \underline{I}(f), \overline{I}(f)\ \exists$, $\Rightarrow \underline{I}(f) \le \overline{I}(f)$

here $\underline{I}(f) := \sup\{\int_a^b s \mid \text{step fun } s \le f\}$

$\overline{I}(f) := \inf\{\int_a^b t \mid \text{stepfun } t \ge f\}$

proof: $\Rightarrow$ see 0.5, 02172

$\Leftarrow$ $\because \underline{I}(f), \overline{I}(f)\ \exists$

$\therefore \{\int_a^b s \mid s \le f\}, \{\int_a^b t \mid t \ge f\}$ nonempty

$\therefore \exists$ step funs s, t, s.t., $s \le f \le t$

$\because s, t$ bounded

$\therefore \exists m, M$, s.t., $m \le s$, $t \le M$

$\therefore m \le f \le M$

f integrable on $[a,b] \Rightarrow f$ bounded on $[a,b]$

proof: $\because f$ integrable $\Leftrightarrow \underline{I}(f) = \overline{I}(f)$ (see 4.3, 02172)

$\therefore f$ bounded.

0.2 set X_1, X_2, Y,

$T: X_1 \to X_2$,

$T_1: X_1 \to Y$,

$T_2: X_2 \to Y$.

if $T_1(x_1) = T_2(T(x_1))$, $\forall x_1 \in X_1$

$\Rightarrow$ ① $T_1(X_1) \subseteq T_2(X_2)$

② if $T^{-1}\ \exists$, then $T_1(X_1) = T_2(X_2)$

proof: ① $\forall x_1 \in X_1$, $\exists x_2 = T(x_1) \in X_2$

s.t., $T_1(x_1) = T_2(T(x_1)) = T_2(x_2)$

$\therefore T_1(x_1) \in T_2(X_2)$

$\therefore T_1(X_1) \subseteq T_2(X_2)$

② if $T^{-1}\ \exists$,

$\forall x_2 \in X_2$, $\exists x_1 = T^{-1}(x_2) \in X_1$,

s.t., $T_1(x_1) = T_2(T(x_1)) = T_2(x_2)$ 0.1.2

$\therefore T_2(x_2) \in T_1(X_1)$

$\therefore T_2(X_2) \subseteq T_1(X_1)$

$\therefore T_2(X_2) = T_1(X_1)$ ▨

0.3 (a) $\{\int_a^b s_1 + \int_a^b s_2 \mid \text{step fun } s_1 \le f, s_2 \le g\}$

$\subseteq \{\int_a^b s \mid \text{step fun } s \le f+g\}$, $\forall a, b \in R$

(b) $\{\int_a^b t_1 + \int_a^b t_2 \mid \text{step fun } t_1 \ge f, t_2 \ge g\}$

$\subseteq \{\int_a^b t \mid \text{step fun } t \ge f+g\}$, $\forall a, b \in R$

proof:

(a) $\forall s_1, s_2$, $s_1 \le f$, $s_2 \le g$.

$\exists s = s_1 + s_2$, $s \le f+g$

$T_1(s_1, s_2) = \int s_1 + \int s_2 = \int(s_1 + s_2) = \int s = T_2(s)$

(note $s = T(s_1, s_2) = s_1 + s_2$ in 0.2)

(here T, T_1, T_2 from 0.2)

(b) The proof is similar. ▨

0.4 (a) $\{s_1 \mid \text{stepfun } s_1 \le cf\}$

$= \{cs \mid \text{stepfun } s \le f\}$, $c > 0$

$= \{ct \mid \text{stepfun } t \ge f\}$, $c < 0$

(b) $\{t_1 \mid \text{stepfun } t_1 \ge cf\}$

$= \{ct \mid \text{stepfun } t \ge f\}$, $c > 0$

$= \{cs \mid \text{stepfun } s \le f\}$, $c < 0$

$\because s(x) \le f(x-c)$ $\therefore s(x+c) \le f(x)$

$\therefore s_1 \le f$

② $\forall$ step fun $s_1 \le f$ on $[a,b]$,

$\exists$ step fun $s(x) = s_1(x-c)$ on $[a+c, b+c]$,

$\therefore s_1(x) = s(x+c)$

$\because s_1(x) \le f(x)$ on $[a,b]$.

$\therefore s_1(x-c) \le f(x-c)$

$\therefore s(x) \le g(x)$ on $[a+c, b+c]$.

③ $\because \int_{a+c}^{b+c} s(x)dx = \int_a^b s(x+c)dx$

$= \int_a^b s_1(x)dx$

$\therefore \{\int_{a+c}^{b+c} s \mid s \le g\} = \{\int_a^b s_1 \mid s_1 \le f\}$

($s_1 = T(s) = s(x+c)$, $s = T^{-1}(s_1) = s_1(x-c)$,

$T_1(s) = \int_{a+c}^{b+c} s = \int_a^b s_1 = T_2(s_1)$

$\therefore$ by 0.2)

(b) similar.

$\boxtimes$

proof: (a)

$T(s) = (s_1, s_2)$,

step fun $s_1 = s|_{[a,c]}$, $s_2 = s|_{[c,b]}$.

$T^{-1}(s_1, s_2) = s$

here $s = \begin{cases} s_1, & \text{if } x \in [a,c) \\ s_2, & \text{if } x \in [c,b] \end{cases}$

$\because T_1(s) = \int_a^b s = \int_a^c s_1 + \int_c^b s_2 = T_2(s_1, s_2)$

$\therefore \{\int_a^b s\} = \{\int_a^c s_1 + \int_c^b s_2\}$ (use 0.2)

(b) similar.

proof: (a)

① if $c > 0$, $\exists T, T^{-1}$

$s = T(s_1) = \frac{1}{c} s_1$, $s_1 = T^{-1}(s) = cs$

$\because T_1(s_1) = s_1 = cs = T_2(s)$

$\therefore \{s_1 \mid s_1 \le cf\} = \{cs \mid s \le f\}$

(use 0.2)

② if $c < 0$, the proof is similar.

(b) similar. $\boxtimes$

0.6 (a) $\{\int_{a+c}^{b+c} s(x)dx \mid$ step fun $s(x) \le g(x) = f(x-c)\}$

$= \{\int_a^b s(x+c) \mid$ step fun $s_1(x) = s(x+c) \le f(x)\}$

here $a < b$.

(b) $\{\int_{a+c}^{b+c} t(x)dx \mid$ step fun $t(x) \ge g(x) = f(x-c)\}$

$= \{\int_a^b t(x+c) \mid$ step fun $t_1(x) = t(x+c) \ge f(x)\}$

here $a < b$

proof: (a) ① $\forall$ step fun $s(x) \le g(x) = f(x-c)$

on $[a+c, b+c]$, $\exists$ step fun $s_1(x) = s(x+c)$

on $[a,b]$,

0.5 (a) $\{\int_a^b s \mid$ step fun $s \le f$ on $[a,b]\}$

$= \{\int_a^c s_1 + \int_c^b s_2 \mid$ step fun $s_1 \le f$ on $[a,c]$, step fun $s_2 \le f$ on $[c,b]\}$

here $a < c < b$

(b) $\{\int_a^b t \mid$ step fun $t \ge f$ on $[a,b]\}$

$= \{\int_a^c t_1 + \int_c^b t_2 \mid$ step fun $t_1 \ge f$ on $[a,c]$, step fun $t_2 \ge f$ on $[c,b]\}$

$\underline{I}(f)+\underline{I}(g) = \sup\{\int s_1 \mid s_1 \le f\} + \sup\{\int s_2 \mid s_2 \le g\}$

$= \sup\{\int s_1 + \int s_2 \mid s_1 \le f,\ s_2 \le g\}$ (see 1,0281)
(additive property of the supremum)

$\le \sup\{\int s \mid s \le f+g\}$ (from 0,3)

$= \underline{I}(f+g)$

Similarly, $\overline{I}(f)+\overline{I}(g) \ge \overline{I}(f+g)$

$\because I(f), I(g)\ \exists$, $\therefore I(f+g)\ \exists$, $I(f+g) = I(f)+I(g)$.

⑤ if $a \ge b$, easy. ▨

2. homogeneous property

$\int_a^b cf(x)dx = c\int_a^b f(x)dx$, $\forall a,b,c \in R$

proof: ① if $a<b$, $c>0$.

$\because \{s_1 \mid s_1 \le cf\} = \{cs \mid s \le f\}$ (from 0,4)

$\therefore \{\int_a^b s_1 \mid s_1 \le cf\} = \{\int_a^b cs \mid s \le f\}$

$\therefore \sup\{\int_a^b s_1 \mid s_1 \le cf\} = \sup\{c\int_a^b s \mid s \le f\}$

$= c\sup\{\int_a^b s \mid s \le f\}$ (from 0281.2)

$\therefore \underline{I}(cf) = c\underline{I}(f)$

$\because T_1(s) = \int_{ka}^{kb} s(x)dx$

$= k\int_a^b s(kx)dx$

$= k\int_a^b s_1(x)dx$

$= T_2(s_1)$

$\therefore \{\int_{ka}^{kb} s \mid s \le g \text{ on } [kb, ka]\}$

$= \{k\int_a^b s_1 \mid s_1 \le f \text{ on } [a,b]\}$ (use 0,2)

The proofs of (a1), (b1), (b2) are similar to that of (a2). ▨

1. additive property

$\int_a^b (f+g) = \int_a^b f + \int_a^b g$, $\forall a,b \in R$

proof: ① if $a<b$,

let $I(f) = \int_a^b f$, $I(g) = \int_a^b g$

$\underline{I}(f) = \sup\{\int_a^b s_1 \mid \text{step fun } s_1 \le f\}$

$\underline{I}(g) = \sup\{\int_a^b s_2 \mid \text{step fun } s_2 \le g\}$

0,7 (a1) $\{\int_{ka}^{kb} s \mid \text{step fun } s \le g = f(\frac{x}{k})$ on $[ka, kb]\}$

$= \{k\int_a^b s_1 \mid \text{step fun } s_1 = s(kx) \le f(x)$ on $[a,b]\}$

if $k>0$, $a<b$.

(a2) change $[ka, kb]$ in (a1) to $[kb, ka]$, if $k<0$, $a<b$.

(b1) $\{\int_{ka}^{kb} t \mid \text{step fun } t \ge g = f(\frac{x}{k})$ on $[ka, kb]\}$

$= \{k\int_a^b t_1 \mid \text{step fun } t_1 = t(kx) \ge f(x)$ on $[a,b]\}$

if $k>0$, $a<b$

(b2) change $[ka, kb]$ in (b1) to $[kb, ka]$
if $k<0$, $a<b$.

proof: We prove (a2).

$\exists T, T^{-1}$, $s_1 = T(s) = s(kx)$,
$s = T^{-1}(s_1) = s_1(\frac{x}{k})$

Similarly, $\overline{I}(cf) = c\overline{I}(f)$

$\because I(f)\exists$, $\therefore I(cf)\exists$, $I(cf) = cI(f)$

② $a<b$, $c<0$.

$\because \{s_1 \mid s_1 \leq cf\} = \{ct \mid t \geq f\}$ (from 0.4)

$\therefore \{\int_a^b s_1 \mid s_1 \leq cf\} = \{\int_a^b ct \mid t \geq f\}$

$\therefore \sup\{\int_a^b s_1 \mid s_1 \leq cf\} = \sup\{c\int_a^b t \mid t \geq f\}$

$= c\inf\{\int_a^b t \mid t \geq f\}$

(from 02181, 2.b)

$\therefore \underline{I}(cf) = c\overline{I}(f)$

Similarly, $\overline{I}(cf) = c\underline{I}(f)$

$\because I(f)\exists$, $\therefore I(cf)\exists$, $I(cf) = cI(f)$.

③ $a \geq b$, easy.

▨

3. linearity property

$$\int_a^b (c_1 f(x) + c_2 g(x))dx = c_1\int_a^b f(x)dx + c_2\int_a^b g(x)dx$$

$\forall a, b, c_1, c_2 \in R$.

3.3 $$\int_a^b \sum_{k=1}^n c_k f_k(x)dx = \sum_{k=1}^n c_k \int_a^b f_k(x)dx$$

$\forall a, b, c_1, \cdots, c_n \in R$

proof: (mathematical induction)

$n=2$, left = right

if $l = r$ for $n-1$,

then, $\int \sum^n c_k f_k$

$= \int (\sum^{n-1} c_k f_k + c_n f_n)$

$= \int \sum^{n-1} c_k f_k + \int c_n f_n$

$= \sum^{n-1} c_k \int f_k + c_n \int f_n$

$= \sum^n c_k \int f_k$

▨

4. comparison theorem

(weak comparison)

$a<b$, f, g integrable, $f \leq g$

$\Rightarrow \int_a^b f(x)dx \leq \int_a^b g(x)dx$

proof: ① method 1.

$\forall s \leq f$, $t \geq g$

$\because f \leq g$ $\therefore s \leq t$

$\therefore \int_a^b s \leq \int_a^b t$

$\therefore \sup\{\int_a^b s \mid s \leq f\} \leq \inf\{\int_a^b t \mid t \geq g\}$

(by 02181.3)

$\therefore \underline{I}(f) \leq \overline{I}(g)$

$\because I(f), I(g)\exists$, $\therefore I(f) \leq I(g)$

② method 2.

$\because f \leq g$

$\therefore \{\int_a^b s_1 \mid \text{step fun } s_1 \leq f\} \subseteq \{\int_a^b s_2 \mid \text{step fun } s_2 \leq g\}$

$\therefore \sup\{\int_a^b s_1 \mid s_1 \leq f\} \leq \sup\{\int_a^b s_2 \mid s_2 \leq g\}$ (by 02181.4)

$\therefore \underline{I}(f) \leq \underline{I}(g)$

$\because I(f), I(g)\exists$ $\therefore I(f) \leq I(g)$

5. additivity with respect to the interval

$\int_a^c f$, $\int_c^b f$ $\exists$, $a, b, c \in R$

$\Rightarrow \int_a^b f \exists$, $\int_a^b f = \int_a^c f + \int_c^b f$

proof: 1) if $a<c<b$.

① $\because \{\int_a^b s \mid s\le f \text{ on } [a,b]\}$

$= \{\int_a^c s_1 + \int_c^b s_2 \mid s_1 \le f \text{ on } [a,c],\ s_2 \le f \text{ on } [c,b]\}$

(from 0.5)

$\therefore \sup\{\int s\} = \sup\{\int s_1 + \int s_2\}$

$= \sup\{\int s_1\} + \sup\{\int s_2\}$

(by 02181.1)

$\therefore \underline{I}(f) = \underline{I}(f,[a,c]) + \underline{I}(f,[c,b])$

② Similarly, $\overline{I}(f) = \overline{I}(f,[a,c]) + \overline{I}(f,[c,b])$

$\because I(f,[a,c]),\ I(f,[c,b])\ \exists$

$\therefore I(f)\ \exists,\ I(f) = \int_a^c f + \int_c^b f$

2) For any other arrangement of a,b,c, the proof is easy.

(see 02163)

6. translation property

$\int_a^b f(x)dx\ \exists \Leftrightarrow \int_{a+c}^{b+c} f(x-c)\,dx\ \exists$.

and $\int_a^b f(x)dx = \int_{a+c}^{b+c} f(x-c)\,dx$, $\forall a,b,c \in R$

proof: 1) if $a<b$.

$\because \{\int_{a+c}^{b+c} s \mid \text{step fun } s(x) \le g(x) = f(x-c)\}$

$= \{\int_a^b s_1 \mid \text{step fun } s_1(x) = s(x+c) \le f(x)\}$

(use 0.6.a)

$\therefore \underline{I}(g) = \sup\{\int_{a+c}^{b+c} s \mid s \le g\}$

$= \sup\{\int_a^b s_1 \mid s_1 \le f\}$

$= \underline{I}(f)$

Similarly, $\overline{I}(g) = \overline{I}(f)$ (use 0.6.b)

$\therefore I(g)\ \exists \Leftrightarrow I(f)\ \exists$, and $I(g) = I(f)$

2) $a \ge b$, easy.

$\blacksquare$

7 expansion or contraction of the interval

$\int_{ka}^{kb} f(\frac{x}{k})dx = k\int_a^b f(x)\,dx$, $a,b \in R$, $k \ne 0$

proof: 1) if $a<b$, $k<0$

① $\because \{\int_{ka}^{kb} s \mid \text{step fun } s \le g = f(\frac{x}{k}) \text{ on } [kb,ka]\}$

$= \{k\int_a^b s_1 \mid \text{step fun } s_1 = s(kx) \le f(x) \text{ on } [a,b]\}$

(from 0.7.a2)

$\therefore \{-\int_{kb}^{ka} s \mid s \le g\} = \{k\int_a^b s_1 \mid s_1 \le f\}$

$\therefore \inf\{\} = \inf\{\}$

$\therefore -\sup\{\int_{kb}^{ka} s \mid s \le g\} = k\sup\{\int_a^b s_1 \mid s_1 \le f\}$

$\therefore -\underline{I}(g) = k\underline{I}(f)$

② $\because \{\int_{ka}^{kb} t \mid t \ge g \text{ on } [kb,ka]\} = \{k\int_a^b t_1 \mid t_1 \ge f \text{ on } [a,b]\}$

(0.7.b2)

$\therefore \{-\int_{kb}^{ka} t \mid t \ge g\} = \{k\int_a^b t_1 \mid t_1 \ge f\}$

$\therefore \sup\{\} = \sup\{\}$

$\therefore -\inf\{\int_{kb}^{ka} t \mid t \ge g\} = k\inf\{\int_a^b t_1 \mid t_1 \ge f\}$

$\therefore -\overline{I}(g) = k\overline{I}(f)$

$\therefore I(g)\ \exists \Leftrightarrow I(f)\ \exists$, $-I(g) = kI(f)$

$\therefore \int_{ka}^{kb} g(x)dx = -\int_{kb}^{ka} g(x)dx = -I(g) = kI(f)$

$= k\int_a^b f(x)dx$

2) if $a<b$, $k>0$, similar.

3) if $a>b$, $k\ne 0$.

$\because -k\int_b^a f(x)dx = -\int_{kb}^{ka} f(\frac{x}{k})dx$

$\therefore k\int_a^b f(x)dx = \int_{ka}^{kb} f(\frac{x}{k})dx$

$\blacksquare$.